

Advanced Programming for HPC

Labwork 2

Students Ho Thanh Thuy Tien - 2540100

1 Implementation

In this labwork, I am using Numba on Google Colab to understand the basic information about GPU.

First of all, I will use `cuda.detect()` and use its id and name to print GPU information:

```
from numba import cuda
cuda.detect()

device = cuda.select_device(0)
print(f"ID: {device.id}")
print(f"Device: {device.name}")
```

Since the CUDA driver does not directly provide the number of cores per multiprocessor, I used a dictionary `cc_cores_per_SM` as a lookup table (that follows NVIDIA's published specifications). The total core count has to be calculated as:

total cores = multiprocessor count × cores per multiprocessor

```
cc = device.compute_capability
sm_count = device.MULTIPROCESSOR_COUNT

cc_cores_per_SM = {
    (2,0) : 32,
    (2,1) : 48,
    (3,0) : 192,
    (3,5) : 192,
    (3,7) : 192,
    (5,0) : 128,
    (5,2) : 128,
    (6,0) : 64,
    (6,1) : 128,
    (7,0) : 64,
    (7,5) : 64,
    (8,0) : 64,
    (8,6) : 128,
    (8,9) : 128,
    (9,0) : 128,
    (10,0) : 128,
    (12,0) : 128
}

core_per_SM = cc_cores_per_SM.get(cc)
print(f"Multiprocessor count: {sm_count}")
print(f"Total cores: {sm_count*core_per_SM}")

meminfo = cuda.current_context().get_memory_info()
print(f"Free Memory: {meminfo[0]} bytes, Total Memory: {meminfo[1]} bytes")
```

2 Result

The information of the GPU is summarized in the following table.

Property	Value
Device name	Tesla T4
Device id	0
Multiprocessor count	40
Cores per multiprocessor	64
Total cores	2560
Total memory	15637086208 bytes ($\approx$ 14.56 GB)
Free memory	15527116800 bytes ($\approx$ 14.46 GB)